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Underbalanced Perforation and Live Well Guns Retrieval in a North Sea nHPHT Well: A Case Study in Multi-Service Provider Integration and Risk Mitigation

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Abstract

Well A, located in a North Sea High-Pressure, High-Temperature (HPHT) prospect, presented unique technical and operational challenges due to its classification as a near High-Pressure, High-Temperature (nHPHT) exploration well. With a prognosed reservoir pressure exceeding 10,000 psi, the well fell outside the conventional design envelope of most regional wells, necessitating a bespoke completion and perforation strategy. This paper presents a case study of the successful execution of underbalanced perforation and live well gun retrieval using Coiled Tubing (CT), highlighting the integration of multiple service providers and the application of advanced project management and risk mitigation practices.

The operation required seamless integration of a coiled tubing deployment system with live well perforation equipment, including a pressure-activated firing head and reverse deployment mechanism. Key challenges included equipment compatibility, pressure integrity, firing sequence control, and safe retrieval of spent guns under live well conditions. These were addressed through structured planning, bi-weekly coordination meetings, a live action tracker, and a full Systems Integration Test (SIT).

A coupled modeling-to-operations workflow was employed, integrating gun-shock analysis, stack Finite Element Analysis (FEA), and inflow performance modeling with a precise field activation script (absolute-pressure firing head, build/hold/bleed sequence), and closing the loop with structured surveillance and history matching. Technical solutions included the use of a J-type circulation sub for fluid displacement, a viscous pill for ball drop assurance, and a water spacer to mitigate fluid incompatibility. The underbalanced design (~700 psi, 11.2 ppg CaCl₂ brine) minimized near-wellbore skin and delivered high early production rates that matched pre-job expectations, with continued Pressure Build-Ups (PBUs) recommended for ongoing reservoir characterization.

Multi-service provider integration was a key enabler, supported by a joint RACI matrix, live action tracker, and early engagement with regulatory authorities. Offshore personnel were included in pre-phase planning, and a full yard SIT ensured robust interface management and operational safety. The operation demonstrated that a properly engineered 10k stack, validated by Finite Element Analysis (FEA) and pressure-tested to 9,750 psi, can host underbalanced perforation and live retrieval without resorting to

heavier 15k systems, reducing handling risk and logistics while preserving well control margins. Zero Non-Productive Time (NPT) and no well control incidents were recorded.

Lessons learned from Well A—including fluid selection, real-time monitoring, contingency planning, and governance—offer a replicable framework for future HPHT completions. This case study establishes new best practices for service sector cooperation and risk mitigation in complex offshore environments, contributing valuable insights for similar high-risk completions.

Introduction

The North Sea HPHT prospect addressed in this paper represents one of the region's more technically demanding developments. Well A was classified as nHPHT exploration well, with a prognosed reservoir pressure exceeding 10,000 psi—well beyond the typical 5,000–6,500 psi design envelope common to many regional wells. The elevated pressure regime, combined with the well's location beneath an existing developed field, created distinctive operational and safety challenges, particularly for the design and execution of the lower completion and perforation program.

The scope of this paper is limited to the operational planning, equipment integration, and execution of underbalanced perforation and live-well gun retrieval using CT in Well A. Subsurface maturation and geological modeling are out of scope. The work is presented as a case study in multi-vendor collaboration, involving the client company, a CT service provider, and a perforation contractor, delivered within a constrained offshore environment.

The main challenge was the safe and effective deployment of perforating guns in a live-well, underbalanced condition using CT. Achieving this required tightly managed interfaces between independent service providers with distinct equipment, operating philosophies, and safety protocols. The CT system–live-well deployment stack interface introduced multiple technical and procedural challenges, including equipment compatibility, pressure integrity, firing head activation, and reverse deployment and recovery of spent guns.

To address these challenges, the project team adopted a structured, collaborative approach. Key enablers included bi-weekly planning meetings, a live action tracker capturing more than 100 actions, and a full SIT at the yard. Interface risks were mitigated through early alignment on the stack-up design, pressure testing of critical components, and a joint RACI matrix to clarify roles and responsibilities. Contingency plans were embedded via pre-approved amendments for credible scenarios such as gun misfire, toolstring retrieval failure, and fluid incompatibility.

This paper documents the technical solutions, project management strategies, and risk-mitigation measures that enabled successful execution of the perforation campaign in Well A. The lessons learned offer practical guidance for integrating multi-service providers in HPHT environments and provide a replicable framework for future operations of similar complexity.

Statement of Theory and Definitions

This section sets out the technical principles and engineering rationale that guided the planning and execution of the perforation campaign in Well A. It summarizes the core theories, design criteria, and operational considerations—including perforation mechanics, charge selection, conveyance options, brine selection, well control strategies, project management principles, well head stress analysis and live-well deployment methodology—that shaped the project's approach.

Perforation Theory

The objective of perforation is to create effective hydraulic communication between the wellbore and the hydrocarbon-bearing formation. In sandstone reservoirs, shaped-charge jets penetrate casing, cement, and rock to form tunnels whose length, diameter, and compaction govern productivity ([Grove](#) og [Manning](#)

2018). Design variables—gun size, charge type, shot phasing, and SPF—are tuned to bypass near-wellbore damage and maximize connectivity, with due regard to reservoir pressure/drawdown and rock strength/grain size.

In HPHT settings, additional risks arise: tunnel collapse, fines migration, and fluid-induced damage. Mitigations include laboratory testing under simulated downhole conditions (e.g., API protocols), numerical modeling (e.g., geomechanics and flow), and, where feasible, real-time monitoring of post-shot response (Ceccarelli, et al. 2010).

- UBP promotes immediate influx, facilitating debris evacuation and minimizing compaction zones.
- OBP can lead to filtrate invasion and low-perm skin unless countered by subsequent stimulation.

Accordingly, effective perforation in sandstone—especially in HPHT wells—requires a holistic approach integrating geomechanics, fluid dynamics, and operational risk management, not merely the mechanical act of creating holes (Satti, et al. 2018).

Charge Type Selection

Charge selection (deep-penetrating vs. big-hole) is driven by formation strength, grain size, damage bypass requirements, and the expected drawdown. Engineers employ laboratory testing (e.g., Section-4/API-style shaped-charge testing), CFD/dynamic event modeling, and downhole condition simulation to compare candidate charges and gun configurations, aiming to maximize penetration while limiting collapse and fines migration (Bonomo og Young 1985). Operational constraints—gun OD, shot phasing, SPF, and conveyance limits—further refine the choice. Evidence from recent case histories shows that optimized deep-penetrating designs and appropriate SPF can deliver material production gains when matched to reservoir conditions (Taggart, et al. 2014).

Conveyance of Perforation Guns

In HPHT wells such as Well A, the choice of Drill Pipe (DP) versus CT has direct implications for well control, gun size, and operational flexibility:

- Drill Pipe (DP)
 - Pros: Accommodates larger/heavier gun systems (e.g., ~4¾-in OD), tight casing alignment, potential for favorable tunnel geometry.
 - Trade-offs: Commonly entails overbalanced conditions to control the well, raising risks of filtrate invasion, skin, and possible need for stimulation. High-density fluids increase losses risk in fragile/depleted zones and complicate barrier envelopes.
- Coiled Tubing (CT)
 - Pros: Enables underbalanced perforation by displacement to lighter, filtered brine (typ. ~8.7–11.2 ppg), improving tunnel clean-up and reducing formation damage; integrates with live-well reverse-deployment to retrieve guns without killing the well. CT with 10,000-psi-class PCE maintains a dynamic barrier envelope; J-type circulation subs and dual checks provide granular pressure control.
 - Trade-offs: Constrains gun OD (typ. ~3-1/8 in), potentially limiting penetration/SPF; introduces risks such as CT collapse under adverse annular pressure. These are mitigated via engineering analysis, pressure testing, and robust string design (Newman, Traugott og Deckert 2009).

Conclusion for Well A. While DP offers mechanical latitude, the well-control and damage risks under HPHT conditions favor a CT-based, underbalanced strategy supported by disciplined barrier management and real-time pressure control (Taggart, et al. 2014).

Brine Type and Costs

A central decision is fluid type at perforation. Industry experience and published guidance favor displacing overbalanced mud with clean, filtered brine prior to UBP to reduce skin and protect the productivity index (Bonomo og Young 1985). Field analogues in the basin have demonstrated that circulating clean and displacing to brine to achieve a static underbalance of ~500–800 psi improves post-shot clean-up and flow stability.

Density selection balances damage, productivity, control, and logistics. Excessive overbalance (e.g., high-density brines deployed for DP UBP) can increase well-control risk and losses. For Well A, completion modeling indicated that acceptable barrier integrity could be maintained across a moderate brine density window (on the order of ~8.7–11 ppg), with the final selection tied to program requirements and pressure-management objectives.

Table 1—Solids Free Brine (Wang 2021)

Chemical Formula	Common Name	Density Range (lb/gal)
ZnBr ₂ –CaBr ₂	Zinc bromide–calcium bromide	15.2–20.5
ZnBr ₂ –CaBr ₂ –CaCl ₂	Zinc bromide–calcium bromide–calcium chloride	15.2–20.5
HCOOK–HCOOCs	Potassium formate–cesium formate	13.1–19.2

Well Control and Barriers in Perforating

With the wellbore displaced to filtered brine, perforation proceeds underbalanced, so fluid is no longer a primary barrier. A dual-barrier philosophy (e.g., NORSOK D-010) is therefore mandatory:

- Primary barrier (example liner case): liner cement and body, liner-top packer, dual-float cemented shoe track.
- Secondary barrier: formation plus cemented strings as applicable, wellhead/XMT seals, and Blow-Out Preventer (BOP) blind/shear rams.

This barrier envelope preserves containment during live-well deployment and reverse retrieval. When non-shearable BHAs are present, clearly identify intervals and apply risk-mitigating procedures throughout deployment. Comprehensive risk assessments, pre-job reviews, and SITs validate readiness. Contingencies include shearable Bottom-Hole Assembly (BHA) options, pre-positioned kill fluid, and real-time pressure surveillance, enabling rapid isolation in the event of excursions or equipment malfunction (Santos og Ribeiro 2017).

Perforation a Long Target Interval Length

To deliver the target interval length efficiently, the team executed single-run conveyance and reverse deployment of the gun string using a live-well reverse-deployment system integrated with CT. Related technologies and workflows are documented in the literature (Samy, et al. 2023). Prior deployments in a sister asset highlighted lessons on stack integrity, connector alignment, and logistics redundancy—including instances where connector engagement issues and BOP components wear required retrieval and re-run. These learnings informed the present campaign's emphasis on:

- SIT of the entire stack-up, including dimensional checks and dummy runs
- Alignment verification across connectors and latching sequences
- Contingency tooling and procedures for misfire, partial unlatch, and retrieval aid (e.g., spears)
- Clear governance via a joint RACI and pre-approved amendments.

Consequently, the selected live-well deployment and reverse-retrieval approach for Well A prioritized reliability, simplified mechanical architecture, and field-proven steps, setting up the detailed procedures and case histories described in the next section (McClean, et al. 2017).

Explanation of Live Well Deployment

Published case histories demonstrate that CT live-well perforation can be executed safely and efficiently without killing the well, thereby minimizing formation damage and enabling optimization of static underbalance in HPHT environments ((McClean, et al. 2017). In the UK Central North Sea, (Taggart, et al. 2014) reported perforating extended intervals under live-well conditions across five HPHT wells; key success factors included pre-job design of CT stretch, circulation pressures, and activation windows.

For HPHT wells, (Satti, et al. 2018) combined laboratory testing (API-style Section-4), CFD, and dynamic event modeling to evaluate shaped charges, wellbore fluid effects, and underbalance levels on tunnel clean-up. Findings included:

- 3-1/8-in systems delivered significant improvements over 2-7/8-in alternatives under representative downhole conditions.
- Wellbore fluid type had minimal impact on charge performance and flow performance given small annular clearance.
- Increasing static underbalance materially reduced tunnel debris and improved clean-up.
- Post-execution productivity exceeded expectations, consistent with the design basis.

Rationale applied to Well A. The operating scenario and deployment methodology are aligned with these case histories. To maximize production efficiency and avoid potential well damage, the client company selected a live-well reverse-deployment system integrated with CT, leveraging the prior art and documented learnings.

Deployment BOP and Stack Adaptation. Previous deployments used a 4-1/16-in lubricator/BOP/valve system. For Well A, stack design and operating requirements necessitated a larger Inner Diameter (ID) deployment BOP compatible with the CT stack-up. The original equipment manufacturer collaborated with engineering to develop new ram inserts and purpose-designed components (e.g., Disconnect, Locator, Pipe/Gripper rams) Fig.1.

Qualification included:

- Standard function tests and pressure integrity checks.
- Pull tests up to 130,000 lbf to validate structural margins.

Given the multi-faceted scope, the program adopted centralized project governance:

- A Perforating Operations Coordinator led execution, supported by live-well deployment specialists trained to host-country regulatory standards.
- Pressure-pumping specialists operated the deployment BOP and manual gate valves.
- Logistics and QA/QC leveraged an onshore ballasting and assembly facility for inspection, redress, functional testing, and final assembly prior to mobilization.

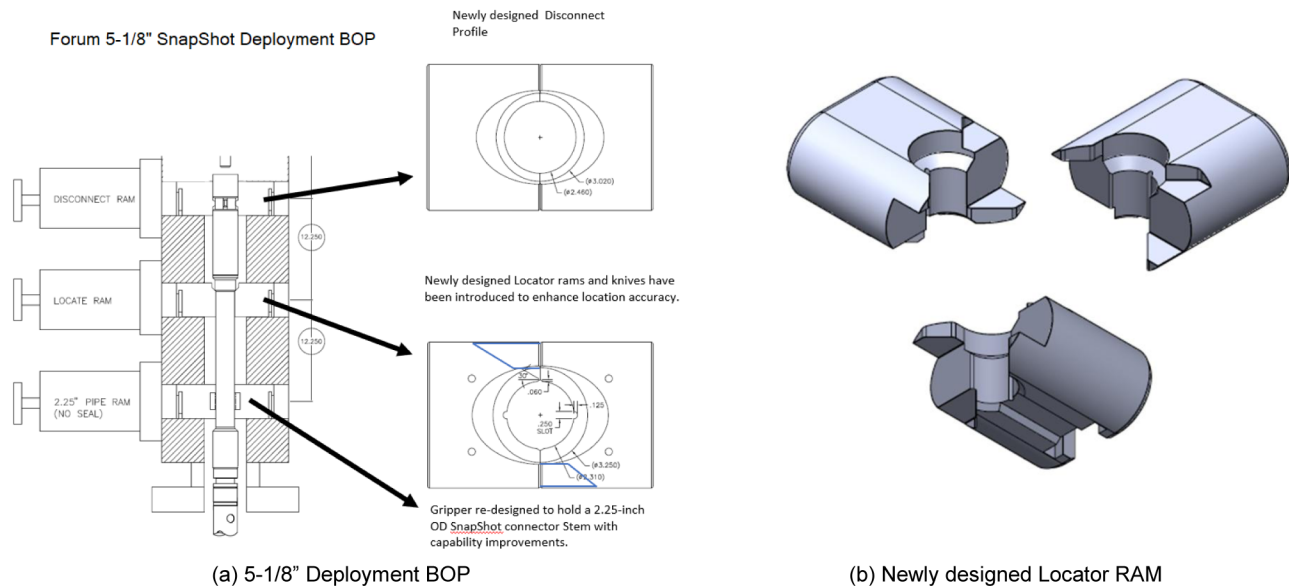


Figure 1—5-1/8" deployment BOP with locator rams

Finite Element Analysis of Well Intervention Stack

Due to the weight and height of the intervention stack, and to ensure safe operations and robust risk mitigation, an engineering assessment was performed using a three-dimensional nonlinear FEA model. The primary goal was to confirm that the intervention stack could safely support coiled tubing weights and operational loads under anticipated field conditions.

The FEA Model simulated dynamic responses, calculating buckling instability, Von Mises stresses, and API-6AF flange bending limits (Newman, Traugott og Deckert 2009). Three main failure modes were evaluated:

- Yielding: Working stress limited to 50% of material yield strength.
- Buckling: Assessed for long, slender stack components under compressive loads.
- Flange Bending: Ensured flange moments did not exceed API-6AF limits, with all bolts meeting recommended makeup stress.

The workflow included characterizing stack components, defining support and constraints, specifying motion patterns, and applying operational loads. Results highlighted any areas exceeding safe stress or bending limits; if found, engineering modifications were made and the model re-run until compliance was achieved.

The results of this analysis are describes in the results section which confirmed that the intervention stack for Well A met industry standards for structural integrity and operational safety in HPHT live-well conditions.

Xmas Tree and Wellhead Design

Selecting appropriate pressure ratings for the Wellhead (WH) and Christmas Tree (XMT) is a critical design decision in HPHT wells, especially when planning live-well operations such as underbalanced perforation and gun retrieval. For Well A, a 5-1/8" 10,000 psi (10k) rated XMT and wellhead system was chosen over a 15,000 psi (15k) alternative, based on technical, operational, and regulatory considerations.

The wellhead and XMT must withstand the Maximum Expected Wellhead Pressure (MEWHP) throughout all operational phases, including perforation, flowback, and potential well control scenarios. For

this campaign, the MEWHP was estimated at 9,250 psi, with a design margin of 500 psi, resulting in a total pressure envelope of 9,750 psi—well within the limits of a 10k-rated system.

This selection aligns with the dual-barrier philosophy required for HPHT integrity. In this configuration, the primary barrier envelope includes the tubing, production packer, and liner top, all pressure tested to 9,750 psi. The secondary barrier comprises the annular space, wellhead seals, and XMT body, also tested to the same pressure. This approach ensures compliance with industry standards such as NORSOK D-010.

Operationally, the 10k system offers several advantages: reduced equipment weight and complexity, simplified rig-up, and minimized handling risks. It also avoids the logistical challenges and costs associated with sourcing and certifying heavier 15k-rated components.

All Pressure Control Equipment (PCE) stack components—including the coiled tubing riser, deployment BOP, and safety head—were rigorously pressure tested to 9,750 psi. Control lines for the Surface-controlled Subsurface Safety Valve (SSSV) were tested to 15,000 psi, ensuring redundancy and compliance with conservative design criteria.

In summary, the decision to use a 5-1/8" 10k XMT and wellhead system for Well A was based on a detailed engineering assessment of pressure regime, barrier integrity, and operational constraints. This provided a technically sound and cost-effective solution for executing underbalanced perforation and live-well gun retrieval in a complex HPHT environment.

The Safety Ram

A safety ram, often referred to as a safety head, is a high-pressure, shearable barrier device installed within the surface PCE) stack during well intervention operations. Its primary function is to provide a final line of defense in the event of unexpected surface pressure excursions or equipment failures. The safety head is designed to sever the coiled tubing or intervention string and securely shut in the well, thereby isolating the wellbore and preventing uncontrolled flow to surface. This capability is especially critical in HPHT environments and live-well operations, where conventional well kill methods may be limited or unavailable. By integrating a safety ram into the PCE stack, operators ensure robust contingency for well control, enhancing operational safety and system resilience throughout complex intervention campaigns.

The surface pressure control envelope for Well A was specified at 10,000 psi, based on a MEWHP of 9,250 psi plus a 500 psi design margin. While the majority of the PCE stack—including the wellhead, risers, deployment BOP, and coiled tubing connector—was rated for 10,000 psi, a 15,000 psi safety ram was incorporated as a critical contingency barrier.

The inclusion of a 15,000 psi safety head (7-1/16") was driven by risk assessment, accounting for scenarios where surface pressures could exceed the 10,000 psi envelope. Uncertainties such as wet gas presence in the reservoir and rapid pressure escalation were considered, with the risk matrix classifying pressures above 9,750 psi as requiring immediate well control intervention.

In such cases, the safety head acts as the final barrier, enabling severance of the coiled tubing and safe shut-in of the well—especially important when the coiled tubing string extends below the SSSV, limiting conventional kill options. Installed above the master valve, the safety head ensures effective isolation and containment.

The decision to implement the 15,000 psi safety head was supported by comprehensive risk analysis, considering failure modes such as SSSV malfunction, tool obstruction, rapid gas influx, and pump failure. The safety head was function and pressure tested to 9,750 psi, with deployment approved through a formal Management of Change (MOC) process.

Description and Application of Equipment and Processes.

The integration of CT operations with the live-well perforation system presented a series of technical and procedural challenges during both planning and execution. These stemmed from the complexity of

coordinating independent service providers—each with proprietary equipment, deployment philosophies, and safety protocols—within aHPHT environment that demanded precise synchronization.

A primary complication involved the deployment stack-up and compatibility of equipment dimensions. The perforating gun BHA, including the firing head and pressure diverter sub, required exact alignment with the CT deployment riser and PCE. Early planning meetings revealed uncertainty regarding the outer diameter of certain components and their ability to pass through the stack-up. This was resolved through a series of dummy runs, dimensional verifications, and a joint review of stack-up drawings, culminating in a successful SIT.

Another significant interface issue was the firing mechanism and ball drop system. The firing head required a specific ball size for activation, while the CT system was originally configured for a different drift size. This mismatch necessitated a redesign of the ball drop sequence and a contingency plan involving a viscous pill to assist ball conveyance. The final procedure included a dummy run to validate the circulation volume required to activate the firing head, ensuring alignment between the hydraulic response of the CT system and the firing mechanism.

Operational sequencing also posed challenges. The need to reverse deploy spent guns required precise coordination to avoid interference with completion equipment. A detailed rig-up sequence was developed in advance, and a RACI chart was reviewed and agreed upon by all parties to clarify roles and responsibilities. Maintaining barrier integrity during reverse deployment led to a revision of the process flow to include a bleed-off line between gate valves, enabling pressure verification without compromising safety.

Communication and procedural alignment were further complicated by differences in documentation and terminology. For example, ambiguity in the dummy run procedure was resolved through joint review sessions, resulting in updated procedures that included a combined dummy run and reverse deployment simulation.

The previous section set out the engineering basis for this campaign: underbalanced perforation to minimize skin; charge/conveyance constraints that favor CT; a dual-barrier well-control philosophy; live-well reverse deployment to avoid a kill; structural assurance via stack FEA; and precise Welltest Loop control to build and bleed pressure in tight firing windows.

This section details how those principles were proven onshore addressing the challenges identifies above and then executed offshore. In brief:

- Barrier philosophy: Full stack pressure tests and function checks before exposure to live well pressures.
- Underbalanced design: A rehearsed pressure program (build to 5,131 psi, bleed to 3,000 psi) to achieve ~500–700 psi underbalance at detonation.
- Live-well deployment: Dummy runs and a reverse-deployment drill to de-risk retrieval under pressure.
- CT conveyance limits: Correlation and spacing strategy (GR/CCL on memory) and careful movement/weight control.
- FEA results: Validated stack configuration, tie-downs and limits carried into the operations envelope.
- Welltest Loop theory: Scripted choke manipulation and communications protocol to hit the 6–7 min firing window.

Equipment Configuration (Overview)

Surface Pressure-Control Envelope (10k).

- Wellhead/XMT and CT PCE including risers, lubricator, in-situ quick-test sub, deployment BOP with reverse-deployment capability, safety head (15k) as contingency.

- All primary components pressure-tested to 9,750 psi prior to live operations; safety head function-tested and included in the stack as final barrier.

Welltest Loop.

- High-pressure choke manifold tied to trip tank/rig separator; provisions to build WHP, bleed down, and monitor returns.
- Pre-job "mock trial" to confirm the pressure-time response needed for the firing-head activation window.

Coiled Tubing (CT) String and BHA.

- CT-conveyed perforating assembly with absolute-pressure firing head (pressure-activated) and reverse-deployment connectors.
- Memory GR/CCL for correlation; venturi junk basket for drift/cleaning.
- Brine displacement to 11.2 ppg CaCl₂ to enable underbalanced perforation and stable returns management.

Onshore Testing and Systems Integration

Stack Pressure Test & Function Checks.

- CT PCE, risers, quick-test sub, deployment BOP, and safety head pressure-tested to 9,750 psi for duration (e.g., 5 min holds).
- Valve function, ram operation, connector latch/unlatch, and instrumentation verified.
- Dimensional verifications.

Dummy Run & Correlation Rehearsal.

- Full-length dummy assembly (guns empty) run through the stack to top shot and to TD; GR/CCL correlations captured to set space-out logic.
- Reverse-deployment drill performed to verify timing, forces, and crew choreography.

Pressure-Sequence Mock Trial.

- With the well tied to the Welltest Loop: pump to 3,000 psi through CT, build annulus to 5,131 psi (1-min hold), then bleed to 3,000 psi within ≤ 7 min.
- All parameters (pressures, times, choke orifice) recorded as the reference for the live firing sequence.

Planned Well Intervention Program

Rig-Up and Initial Validation.

- R/U CT PCE, risers, and deployment equipment; pressure test to 9,750 psi.
- Line up Welltest Loop for returns management.
- Retrieve shallow set devices on CT as required for access and barrier verification.

Drift, Clean-Out, and Brine Displacement.

- Drift to TD using a 3.625-in venturi junk basket; circulate at controlled rates.

- Displace to 11.2 ppg CaCl₂ brine, using a fresh-water spacer (6–10 bbl) to avoid incompatibility; maintain clean returns.
- Circulation test across the choke to confirm the loop response and stable flow control.

Dummy Gun & Correlation Run (with Reverse-Deployment Test).

- Run dummy guns with GR/CCL to target depth; correlate and flag CT depth.
- Test reverse deployment of the lower dummy section through the deployment BOP; confirm timings and procedure steps.
- Crew alignment: roles/responsibilities reinforced; communications paths among CT control, pumps, and Welltest choke verified.

Make-Up and Deployment of Live Guns.

- Make-up perforating BHA; adhere to exclusion zones and handling controls for energetic components.
- Deploy on CT and space-out across the interval based on GR/CCL and flagged CT.
- Observe speed limits (e.g., 30–50 ft/min, 10 ft/min through restrictions) and weight checks to keep within modeled limits.

Firing Sequence—Pressure Build and Bleed (Underbalanced Perforation).

- Pressurize via CT to 3,000 psi against the choke.
- Build annulus to 5,131 psi (1-min hold) to initiate the absolute-pressure firing head (estimated ~6–7 min delay).
- Bleed down through the choke to 3,000 psi within ≤ 7 min, establishing ~500–700 psi underbalance at detonation.
- Monitor detonation indications: WHP response, CT weight/torque signatures, and acoustic/pressure transients.

Reverse Deployment and Recovery.

- On confirmation of firing, POOH and reverse-deploy spent guns using the live-well deployment system.
- Maintain pressure-control discipline during retrieval; watch for trapped gas in gun spacers; follow connector pull-test practices.

Close-Out and Handover.

- R/D CT and install tree cap; pressure test to 9,750 psi.
- Complete well handover to production after integrity checks and documentation review.

Presentation of Data and Results

This section presents the integrated technical results that underpin the project's engineering design and operational strategy. By combining predictive modeling, equipment qualification, and field surveillance, we demonstrate how theoretical planning was translated into measurable field performance. The following results encompass well performance modeling, gun-shock and stack integrity analysis, pressure control system validation, and post-perforation production monitoring. Where direct quantitative outputs are unavailable, qualitative assessments and operational observations are provided to ensure a comprehensive

and transparent evaluation of system performance. Together, these data sets offer a holistic view of how the design basis was realized in practice, supporting both the safety and productivity objectives of the campaign.

Gun-Shock Modeling and Field Verification

Modeling Objective. A conceptual gun-shock analysis was conducted to verify the mechanical integrity of the bottomhole assembly (BHA), connector interfaces, and surface stack in response to dynamic loads generated during perforating gun detonation. Acceptance criteria were established to ensure that all critical components remained within their mechanical limits and that post-shot operability—particularly for reverse deployment—was maintained throughout the operation.

Field Verification. Operational results were fully consistent with the modeling predictions (Fig. 2). All perforating guns fired as planned, with post-job inspection confirming the absence of misfires or mechanical failures. The gun string was successfully reverse-deployed under pressure, and no barrier non-conformances were reported at handover.

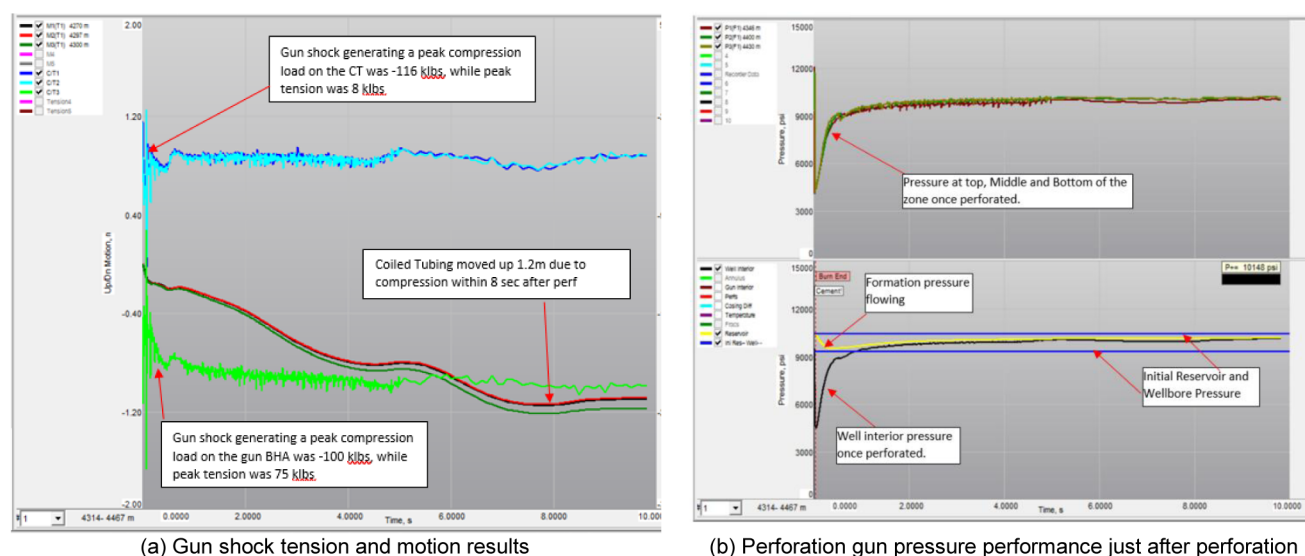


Figure 2—Modeled versus observed gun-shock and connector response

Finite Element Analysis of Well Intervention Stack Results

Analytical Method. A three-dimensional nonlinear FEA, was performed on the intervention stack to evaluate its structural integrity under both dynamic and operational loading conditions. The assessment included buckling stability, component stress distribution (Von Mises criteria), and API-6AF flange bending performance. Dynamic excitation from wellhead and platform motions, as well as operational loads such as coiled tubing weight, tension, and wellhead pressure, were incorporated into the model. Acceptance criteria required that component stresses remain below 50% of material yield strength and that flange moments not exceed 100% of the API-6AF limit, with all bolts made up to the recommended Make-Up Stress (MUS), as outlined in the theory section.

Results. The as-configured stack satisfied all structural criteria across the range of modeled scenarios. In cases where sensitivities approached design limits, iterative adjustments were made to lubricator size, tie-downs, and flex-element selection until full compliance was achieved. The final configuration was then locked for field execution. No deviations or non-conformances were observed during rig-up pressure testing, confirming the validity of the analytical predictions (Fig. 3 (a), (b)(c)).

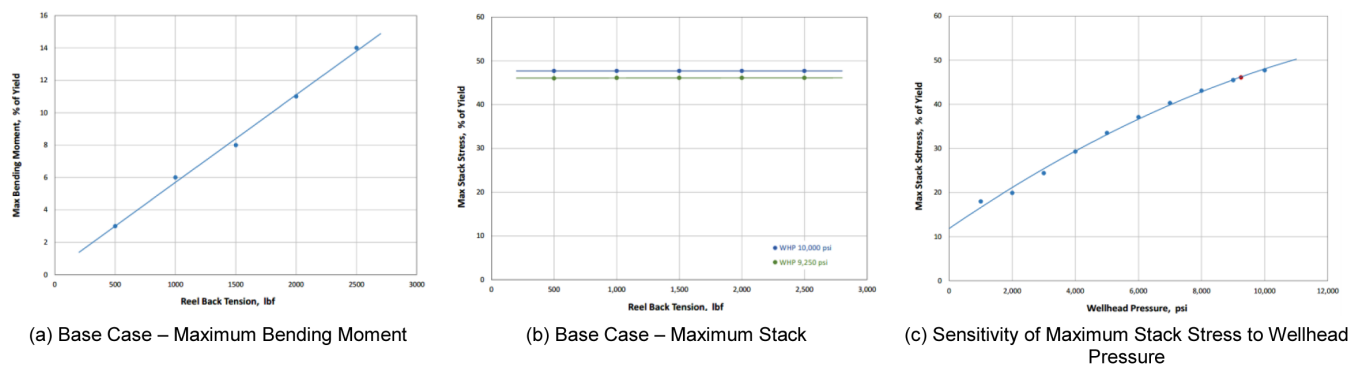


Figure 3—FEA analysis utilization plots—component stress versus yield and flange moment versus API-6AF limit

Pressure Control Equipment Stack and Deck Layout

The successful outcome of the surface-stack FEA provided the necessary assurance to proceed with the rig-up of the well intervention PCE stack. The as-built surface envelope was configured for 10,000 psi, with a 15,000 psi safety head included as a contingency barrier.

Prior to live operations, all critical components—including the wellhead/XMT, coiled tubing PCE, risers, lubricator, deployment BOP, and safety head—were subjected to comprehensive function and pressure testing at 9,750 psi, in accordance with the design basis. The safety head was retained in the stack to address any potential over-envelope scenarios.

Completion and rig-up documentation confirm that the 10k CT stack was installed as planned, and all primary and secondary barriers were tested per program requirements (Fig. 4 (a, b, c)). No failures or non-conformances were observed at handover, validating both the analytical predictions and the operational readiness of the system.

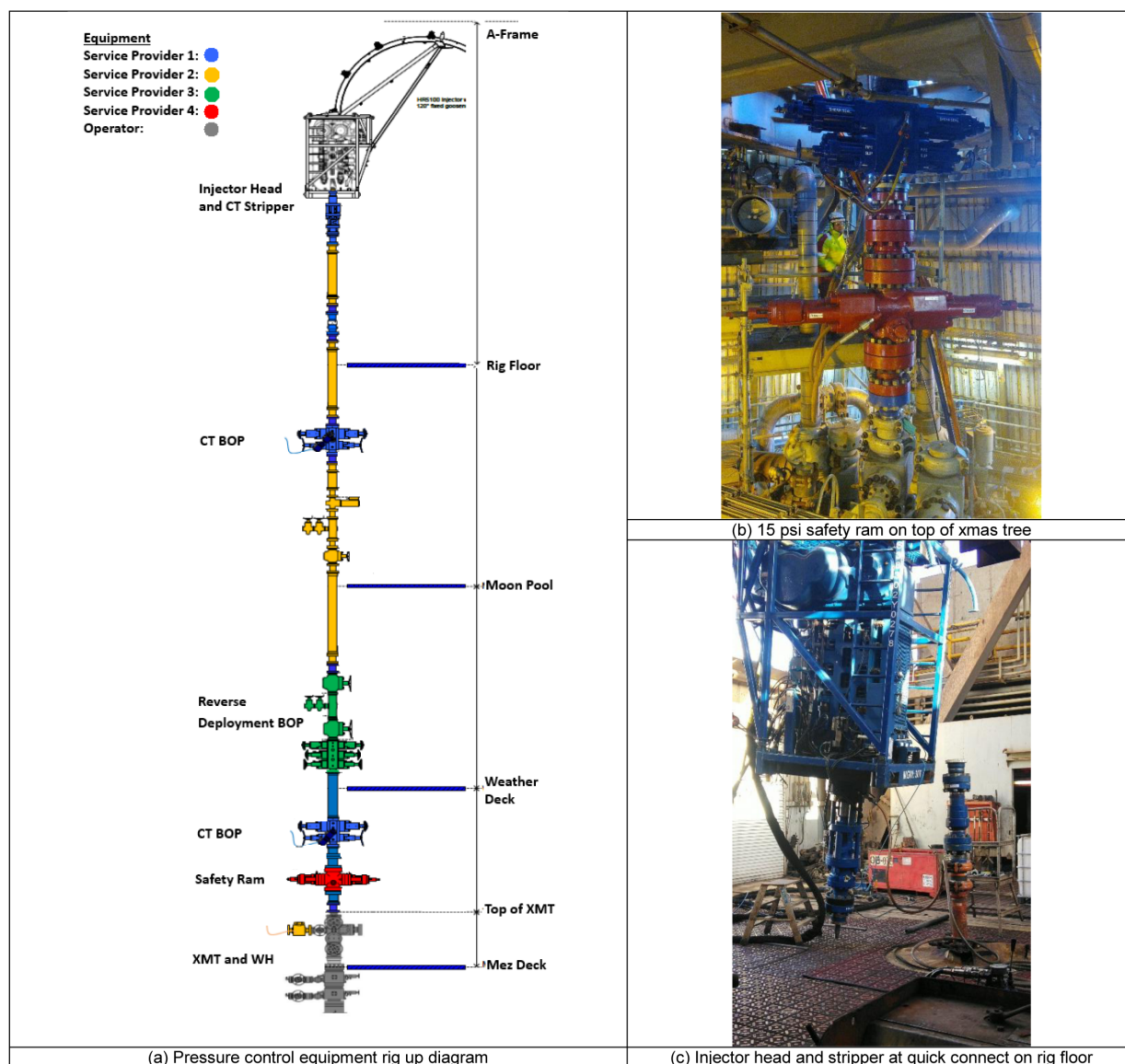


Figure 4—Coiled tubing multiservice provider rig up and interface

Pressure Control Equipment Stack and Deck Layout

The successful completion of the PCE rig-up and pressure testing, underpinned by positive FEA results, enabled the intervention team to proceed with the perforation operation as planned. Following wireline logging and correlation, the perforating gun configuration was re-evaluated and reconfigured to match the as-logged interval across the formation. This resulted in a revised primary gun loading using 3-1/8" guns, with a backup 2-7/8" system prepared for contingency. The updated perforation design also necessitated a revision of the end-of-life well performance model to reflect the actual interval and gun loading deployed.

Configuration and Deployment. The primary run utilized 3-1/8" guns, spaced out according to gamma ray and CCL correlation, and loaded to match the logged reservoir interval. The absolute-pressure firing head was selected to enable live-well operations and support reverse deployment. The gun system summary, including OD, Shots Per Foot (SPF), and phasing, is provided in Fig. 5 (a) & (b).

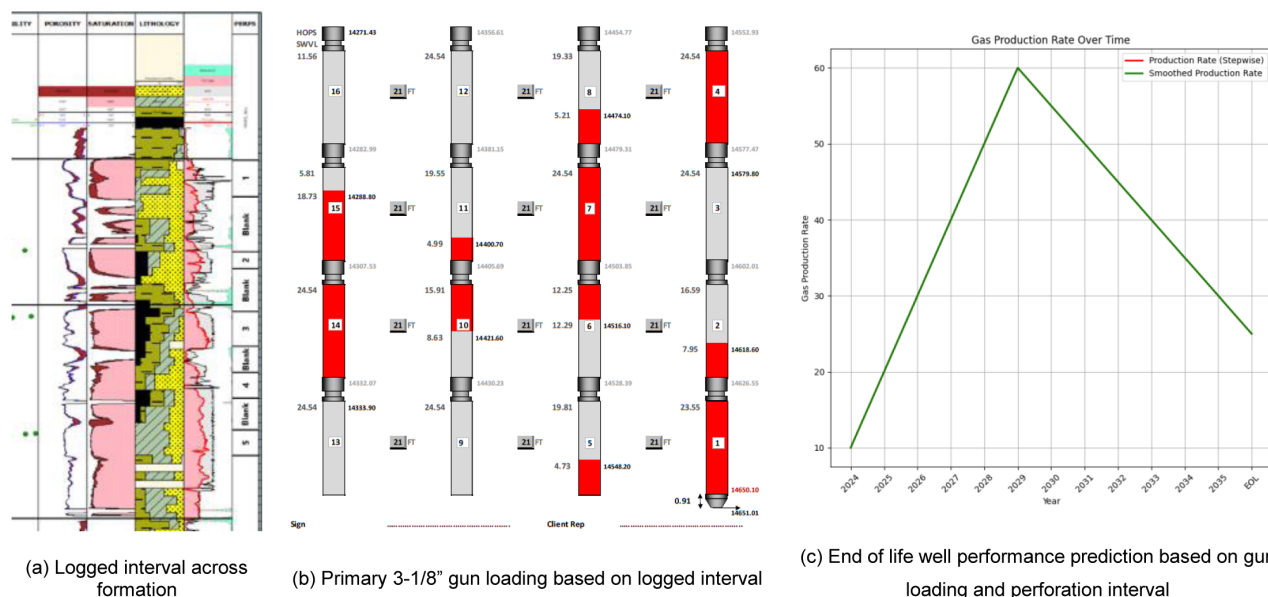


Figure 5—Gun loading and well performance prediction based on logged interval

Activation and Pressure Program. The activation sequence was executed as rehearsed: annulus pressure was built to the firing threshold and held for one minute, followed by a controlled bleed-down to the underbalance set-point within the firing head's delay window. The operation was performed underbalanced (~500–700 psi) to promote effective tunnel cleanup and minimize near-wellbore skin. All guns fired as planned, and the spent gun string was reverse-deployed under pressure. Post-job inspection confirmed the integrity of all components and the absence of misfires.

Perforation Execution and Gun System Optimization. Following successful rig-up and pressure testing of the PCE stack, the perforation operation was executed as planned. After logging, the gun configuration was re-evaluated and reconfigured to match the as-logged interval, resulting in a revised primary 3-1/8" gun loading. This adjustment required a corresponding update to the end-of-life well performance model (Fig 5 (c)). The updated model, based on the as-built perforation design, confirmed that the well would maintain high deliverability and a low water-gas ratio throughout its productive life. The activation sequence—building annulus pressure to the firing threshold, holding for one minute, and bleeding down to the underbalance set-point—was performed underbalanced (~500–700 psi), promoting effective tunnel cleanup and minimizing skin. All guns fired as planned, and the spent gun string was reverse-deployed under pressure. Post-job inspection confirmed the integrity of all components and the absence of misfires.

The Inflow Performance Relationship (IPR) and well performance modeling for Well A were designed to optimize initial productivity and minimize near-wellbore damage through underbalanced perforation and early drawdown. The modeling workflow was iterative, incorporating pre-job predictions, post-startup surveillance, and ongoing history matching. Key milestones included the initial Completion Test Report (CTR) in early January, structured history matching during Q1–Q2, Pressure–Volume–Temperature (PVT) laboratory experiments in June, and a subsequent fluid model update. This process ensured continuous calibration between measured rates, pressures, and model outputs. The impact of shot density and charge selection on well productivity was also evaluated: while analogue well was perforated using 6 spf and other affiliate wells used 4 spf, modeling and field experience demonstrated that optimal gun system selection—including deep-penetrating charges and higher shot density—can yield up to a 20% increase in production (Grove og Manning 2018). [MA5]This enhancement was confirmed by perforation performance modeling, which showed that a 12 spf density can deliver significantly higher productivity compared to 4

and 6 spf (Ceccarelli, et al. 2010), as illustrated in Fig. 6. These findings are consistent with the observations of (Samy, et al. 2023), who reported similar productivity gains through optimized perforation design.

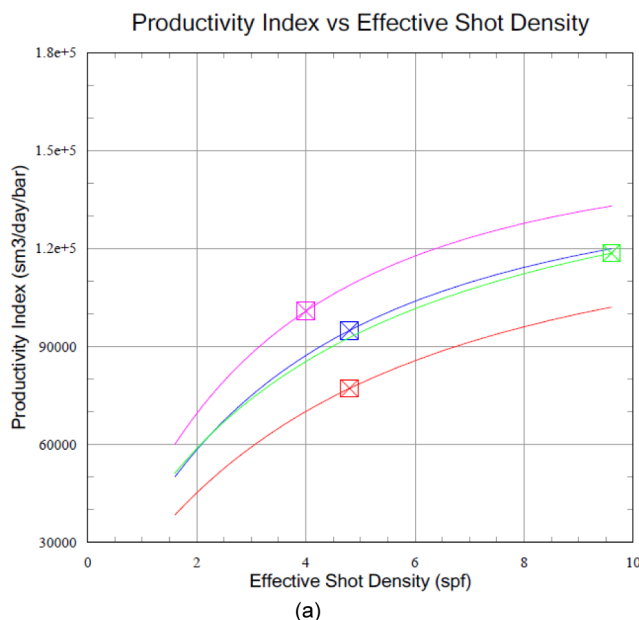


Figure 6—PI vs Shot Density for Various Charges

Observed Production Performance. Following perforation and handover, Well A stabilized at approximately 60 Mscfpd of gas (~11 kboe/d) and ~5 kbopd of condensate, yielding a combined rate of ~16 kboe/d (Fig. 7). The Condensate-Gas Ratio (CGR) averaged 85 stb/Mscf. Facility records indicate a high-deliverability gas-condensate system and low initial skin. December production allocations carry minor uncertainty due to ongoing calibration of test-separator and meter factors.

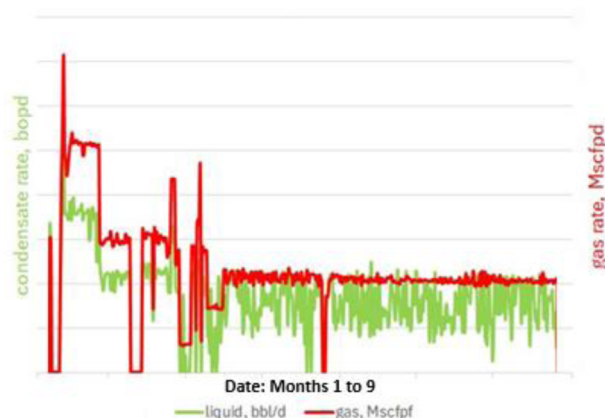


Figure 7—Well production rates 9 month after perforation indication a successful job

Fluid Characterization and Flow Assurance. Early production behavior and PVT analyses confirm a gas-condensate system, closely analogous to neighboring fields. Initial log interpretations suggesting a possible oil rim are not supported by current production data. Pour-point and wax-inhibitor testing, as well as H₂S monitoring during clean-up and stabilized flow, indicate moderate and manageable flow-assurance risk.

Operational Validation and Stability. The underbalanced perforation (~700 psi) on a 10k CT rig-up was executed without incident; all guns fired, reverse deployment was completed, and all barriers were tested per program with no failures at handover. The start-up and sampling record notes some instability in the

receiving network during the same timeline, which impacted test windows and logistics—these factors are accounted for in the data quality statements.

Implications for IPR and Future Surveillance. The observed rates and stable cleanup under underbalanced conditions are consistent with pre-job expectations for low near-wellbore skin. Continued structured PBUs and well-test mini-build-ups are recommended to further refine Productivity Index (PI), skin, and reservoir connectivity as the well matures (Fig. 8).

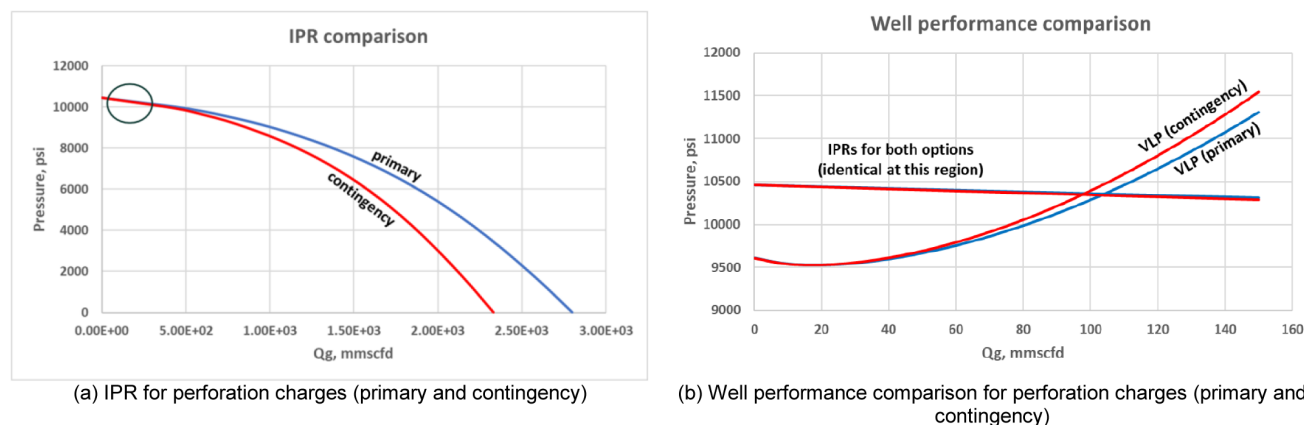


Figure 8—IPR and well performance comparisons for primary and secondary charges.

Conclusions

This case study demonstrates, for the first time in this basin, a fully integrated live-well reverse gun retrieval on CT under nHPHT conditions using a 10k surface envelope complemented by a 15k safety head as contingency—an architecture validated by FEA and proven in the field. By engineering the 10k stack to a 9,750 psi test level and governing operations with a strict dual-barrier philosophy, the team robustly executed underbalanced perforation and live retrieval without resorting to a heavier 15k system, thereby reducing handling risk and logistics while preserving well-control margins. The activation choreography—absolute-pressure firing head with a build/hold/bleed script—landed a ~500–700 psi static underbalance at detonation and set the foundation for a low-skin cleanup.

A coupled modeling-to-operations workflow tied gun-shock analysis, stack FEA, and IPR/performance modeling directly to that field activation script and then closed the loop with surveillance and history matching. The outcome was clear model–field alignment: observed rates and CGR fell within the pre-job envelope once allocation caveats were applied, with combined early production near ~16 kboe/d confirming that the perforation system and drawdown strategy delivered the intended cleanup and productivity. Ongoing PBUs and mini-build-ups will tighten the characterization of IPR, skin, and connectivity as the well moves from cleanup into steady-state operations.

System integrity under dynamic loads matched the analytical acceptance limits: there were no misfires, no barrier exceptions, and the live-well reverse-deployment sequence ran cleanly. Field results corroborated the API-6AF flange-bending checks and the $\leq 50\%$ yield utilization targets established in the FEA work, underscoring the value of combining analytical margins with rehearsed field procedures before exposing the stack to live pressure. Operationally, the job ran underbalanced (~700 psi) on a 10k CT rig-up with successful reverse deployment and a clean handover; while receiving-network instabilities constrained some test windows during start-up/sampling, these effects were captured in the data-quality statements and are being addressed through allocation hygiene (final PVT constants and meter-factor calibration, with re-allocation as needed).

Equally important, the paper establishes multi-service provider integration as a deliberate engineering control. Beyond the technical selections—3-1/8" guns, memory GR/CCL correlation, and brine displacement with spacer—the governance layer (RACI, live action tracker, pre-approved amendments) and a yard Systems Integration Test functioned as risk-reduction mechanisms on par with hardware choices. Three reproducible steps underpinned repeatability: a mock trial of the pressure-time sequence on the Welltest Loop; a combined dummy run plus reverse-deployment rehearsal to de-risk retrieval under pressure; and dual-barrier discipline with a 10k PCE envelope and 15k safety head as the final contingency. Together, these created a high-fidelity rehearsal of the critical path and stabilized cross-vendor interfaces offshore.

For practitioners, the consolidated takeaway is that underbalance plus immediate drawdown offers a practical, low-skin pathway in HPHT sandstone when coupled with disciplined barrier management and reverse retrieval, avoiding the cleanup penalties of overbalance or kills. Just as critically, the up-front integration work—SIT, dimensional checks, RACI, and activation mock trials—must be treated as hard engineering gates, not optional project-management artifacts. In sum, the novelty lies in (i) the first integrated demonstration of CT live-well reverse retrieval at nHPHT on a validated 10k/15k-contingency surface system; (ii) the end-to-end, traceable modeling-to-operations loop from gun-shock/FEA through surveillance; and (iii) the codified multi-vendor governance that turns a complex HPHT perforation into a repeatable playbook for future campaigns.

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Nomenclature

API-6AF	Flange bending assessment methodology referenced for acceptance limits in well intervention stacks (per API 6A/6AF context)
API Section-4	Laboratory perforating charge test protocol (API-style Section 4) for downhole condition simulation.
BHA	Bottomhole Assembly.
BOP	Blowout Preventer (deployment BOP in the live-well CT stack).
bbbl	Barrel (volume unit).
CaBr₂	Calcium bromide (brine component).
CaCl₂	Calcium chloride (brine component).
CCL	Casing Collar Locator (depth correlation tool).
CGR	Condensate–Gas Ratio (typically in stb/Mscf).
CT	Coiled Tubing.
DP	Drill Pipe.
FEA	Finite Element Analysis (nonlinear structural modeling of the intervention stack).
ft/min	Feet per minute (movement speed).
GR	Gamma Ray (log) used for depth correlation.
H₂S	Hydrogen sulfide.
HPHT	High-Pressure, High-Temperature.
IH	Injector Head (CT injector assembly).
IPR	Inflow Performance Relationship.
ID	Inner Diameter.
kbopd	Thousand barrels of oil per day.
kboe/d	Thousand barrels of oil equivalent per day.

lbf	Pound-force.
MEWHP	Maximum Expected Wellhead Pressure.
Mscf/d	Thousand standard cubic feet per day (also seen as Mscfpd).
MUS	Makeup stress (recommended bolt makeup stress in flange analysis).
nHPHT	Near-HPHT (near high-pressure, high-temperature classification).
NORSOK D-010	Norwegian standard for well integrity (barrier philosophy).
OBP	Overbalanced perforation.
OD	Outside Diameter.
PBU	Pressure Build-Ups (surveillance tests).
PCE	Pressure Control Equipment (surface stack: risers, lubricator, deployment BOP, etc.).
PI	Productivity Index.
POOH	Pull Out Of Hole.
ppg	Pounds per gallon (fluid density unit).
psi	Pounds per square inch (pressure unit).
PVT	Pressure–Volume–Temperature (fluid property analysis).
RACI	Responsible, Accountable, Consulted, Informed (governance matrix).
R/D	Rig Down.
R/U	Rig Up.
SIT	Systems Integration Test (yard/onsite rehearsal and validation).
SPF	Shots per foot (perforation density).
SSSV	Subsurface Safety Valve.
stb	Stock-tank barrel.
TD	Total Depth.
UBP	Underbalanced perforation.
WHP	Wellhead Pressure.
WH / XMT	Wellhead / Christmas Tree.
ZnBr₂	Zinc bromide (brine component).
HCOOK	Potassium formate (brine component).
HCOOCs	Cesium formate (brine component).

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